

BS EN 17206:2020 Mechanical Inspection Extract

Entertainment technology - Machinery for stages and other production areas - Safety requirements and inspections

Prepared for Stage Electrics

Source reviewed: BS EN 17206:2020 incorporating corrigendum October 2021. This extract summarises the information most relevant to mechanical inspections, integrated lifting systems, stage machinery, lifting accessories, load bearing equipment, safety devices, documentation, testing and commissioning. It is not a substitute for the full standard, LOLER, PUWER, manufacturer instructions, or a competent person's judgement.

Executive summary

- BS EN 17206 applies specifically to stage machinery and production-area machinery, including guided and unguided loads, control systems, electrical/electronic systems, hydraulic and pneumatic systems, and machinery used to move performers during artistic performances.
- The standard is directly relevant to theatre and entertainment installations such as fly bars, lighting bars, point hoists, performer flying systems, stage and orchestra elevators, revolving stages, stage wagons, trap elevators and related machinery.
- For inspection purposes, the standard is strongest in three areas: design and load-bearing requirements, safeguarding and safety functions, and testing prior to first use or after changes.
- The document uses the Entertainment Load Limit (ELL) rather than Safe Working Load terminology. Where the machine has a different permissible load at rest, it uses ELL/R.
- A competent person is required for intended use definition and risk assessment, and an examiner determines the type and extent of testing after changes where applicable.
- Documentation is a major inspection requirement: technical data, markings, operating manuals, installation instructions, maintenance instructions, inspection/examination guidance, test certificates and test logs are all expected.
- The standard does not replace statutory LOLER requirements. It should be used alongside LOLER thorough examination duties and manufacturer information.

1. Scope and relevance to Stage Electrics

The standard covers machinery, machinery installations and control systems used in places of assembly and staging or production facilities. Examples include theatres, multi-purpose halls, exhibition halls, film, TV and radio studios, concert halls, schools, bars, discotheques, open-air stages and other rooms for shows and events. It applies to machinery installations with guided or unguided loads and covers machinery that includes controls, electrical and electronic control systems, hydraulic and pneumatic power supplies.

Relevant equipment / installation	Inspection relevance
Fly bar systems and lighting bars	Inspect as load carrying devices and associated load bearing equipment, drive systems, terminations, safety functions, ELL marking and loading notices.
Point hoists and chain hoists	Check load bearing lines, hooks, brakes, safety devices, overload/underload protection where applicable, ELL marking and test records.
Performer flying systems	Higher consequence use case: check safety functions, redundancy, speed/position monitoring, emergency stop, rescue/recovery provision and documentation.
Stage elevators, orchestra elevators and trap elevators	Check load assumptions, protective spaces, blocking devices, shaft/landing doors, interlocks, fall protection, ELL/R, emergency stops and load testing evidence.
Revolving stages, turntables and stage wagons	Check drive/brake function, travel hazards, collision prevention, guarding/fall protection, testing after changes and control system limits.
Integrated lifting systems / multi-machine systems	Check group synchronisation, collision prevention, structural overload protection, control logic, simultaneous movement limits and test evidence.

2. Key terms to use in inspection reports

Term	Meaning / inspection significance
Competent person	A person with enough practical and theoretical knowledge and experience to carry out their duties, and who understands the limits of their competence.

Stage Electrics - BS EN 17206:2020 Mechanical Inspection Extract

Machinery installation	All elements between the load and the machinery anchor point. This is the closest term to an integrated lifting system.
Load bearing equipment	Assembly of load bearing elements, including the drive mechanism.
Load bearing element	Parts of a machine between the load and the machinery anchor point.
Load carrying device	The part directly carrying the intended load, such as a fly bar, elevator platform, truss or point hoist hook.
Lifting accessory	Component or equipment placed between lifting machinery and the load, placed on the load, or forming an integral part of the load to allow the load to be held.
Entertainment Load Limit (ELL)	Maximum load that an item of lifting equipment is designed to raise, lower or sustain.
Entertainment Load Limit at Rest (ELL/R)	Maximum load the equipment is designed to sustain at rest; may differ from the moving ELL.
System load	ELL plus the weight of the load carrying device.
Characteristic load	System load plus dynamic forces during normal operation, including holding loads at rest.
Load at failure	System load plus dynamic forces occurring due to uncontrolled stops in a failure condition.
Test load	Load used when testing a lifting device, load bearing equipment, or load carrying or securing devices.

3. Hazards that should influence inspection scope

The standard requires foreseeable hazards to be identified and risk assessed. The following are particularly relevant to mechanical inspection activities in theatres, venues and production spaces:

Hazard group	Examples to consider during inspection
Mechanical hazards	Falling objects, gravity, instability, kinetic energy, rotating elements, sharp edges, inadequate strength, unsuitable ropes/chains/accessories and failure of drive or load bearing components.
Lifting-specific hazards	Overload, overturning moments, unexpected movement of loads, inadequate holding devices, unsuitable accessories, collision between machines, incorrect installation, testing, use or maintenance.
Movement of persons	Insufficient mechanical strength, control system malfunction, software errors, uncontrolled movement, overspeed and loss of synchronisation.
Environment	Moisture, water, temperature, electromagnetic disturbance, low visibility, fog/dust effects, restricted access and local lighting.
Human factors	Control-device identification, unauthorised start-up/use, visibility from control positions, operator competence and insufficient instructions.

4. Load-bearing and mechanical inspection considerations

4.1 General design concept

- Stage machinery is grouped into load bearing equipment, control systems, drive systems and load carrying devices.
- The basic safety concept is intrinsic safety or single-fault safety, achieved either by doubling the working coefficient in calculations or by redundancy.
- Load bearing elements are generally designed for twice the characteristic load. Failure-condition design considers the load at failure.
- Load bearing structural elements should use appropriate structural standards; welding of steel/aluminium fabrications is expected to follow EN 1090-2 or EN 1090-3.
- Natural frequency, lateral torsional buckling and lateral buckling should be considered.

4.2 Wire ropes, chains and load bearing lines

Item	Inspection points
Wire ropes	Check specification, condition, route, fleet angles, ability to inspect full length, terminations and evidence of type 2.2 inspection documentation. Jacketed wire ropes are not to be used as part of the machinery installation.
Wire rope safety factors	For load bearing lines, the standard refers to at least 10 at characteristic load and at least 2 at load at failure. Lower factors may only be justified by risk assessment where ropes do not

	travel over pulleys or drums.
Chains	Round steel chains in machinery installations are to be calibrated and tested to EN 818-7, with a safety factor of at least 8 at characteristic loading and no permanent deformation at load at failure.
Chain documents	Chains should have type 2.2 inspection documentation confirming testing. Other chain types need analogous treatment, taking type-specific characteristics into account.
Unintentional loosening	All load bearing elements should be secured against unintentional loosening; quick links are given as an example where orientation matters.
Materials	Structural elements of load bearing equipment should be non-flammable. Synthetic or natural fibre ropes may only be used where the risk assessment shows it is safe.

4.3 Terminations, hooks, drums and pulleys

Area	What to check
Wire rope terminations	Terminations should retain at least 80% of the rope minimum breaking force and be inspectable. End connections should meet EN 13411. Grips are not allowed as the means to terminate load bearing lines.
Detachable terminations	Examples include asymmetric and symmetric wedge sockets. Detachable terminations should be secured against self-detachment.
Fixed terminations	Examples include splices, ferrules and swage terminals. Verify compatibility with the installation and documentation.
Load hooks	Hooks should prevent unintentional detachment of the load, by safety device or hook shape. Hooks with safety latches meet this requirement.
Chain end connections	Steel chain end terminations should withstand 100% of the chain minimum breaking force before failure. Threaded connections should be locked and verifiable.
Wire rope drums	Round rope drums should be helically grooved. Rope crossovers and improper winding should be prevented or detected. Drum diameter should not be less than 18 times rope diameter.
Diverter pulleys	For round wire ropes, pulley diameter should be at least 20 times rope diameter and groove geometry should suit the rope.
Chain sprockets	Chain drives should prevent jumping out, twisting and jamming.

4.4 Drive systems and load securing devices

- Drive systems are required to preclude unintentional hazardous movements.
- This may be achieved by movement-operated load securing, such as a dynamically self-sustaining gearbox, or by at least two independently functioning load securing devices such as independent brakes.
- The load securing devices should be capable of bringing the test load to rest even if one device fails, and it should be possible to check each device separately at rated speed.
- Where multiple brakes, hydraulic valves or clamps are used, they should be engaged by at least two independent devices.
- Risk assessment may justify one load securing device in limited cases, for example some group drive arrangements or certain horizontal movements.
- Couplings, friction couplings, switchable couplings, brakes and springs have specific design requirements; inspection should verify any manufacturer evidence and functional test method.
- Guided power-driven suspended load carrying devices should have a device that interrupts movement when the suspension system becomes slack.

5. Safeguarding and access requirements relevant to inspection

Area	Inspection focus
Protective spaces under stage machinery	Where inspection/maintenance occurs under machinery, protective spaces or blocking devices are required. Stage elevators should be lowerable only to a safe protective space or supported by a suitable blocking device.
Blocking devices	Should support the load carrying device, be operable/installed by one person from a safe position, be accessible and either fitted permanently or stowed nearby.
Access openings	Openings to protective spaces should be at least 0.6 m x 0.8 m. Fixed lighting is required in maintenance areas under stage elevator platforms.
Walkways and maintenance areas	Maintenance walkways between drives and switchgear should be at least 0.6 m wide with at least 2 m clearance height where used during maintenance or monitoring.
Crushing, shearing and trapping	These points are to be avoided or guarded. If guarding is not practicable, protective

	measures such as clearance, full view of travel area and hold-to-run controls should be defined.
Stage floor gaps	Gaps caused by moving stage elements should not exceed 20 mm unless suitable protective means are provided.
Shaft walls and landing doors	Doors should be smooth/flush on the shaft side, interlocked, closed and locked before movement, and only open when the platform is stationary within 0.2 m of the landing.
Counterweights	Cradles should prevent weights falling out. Counterweight tracks should be covered/guarded, and loading/unloading should be possible without entering the hazard zone where required.

6. Safety functions, controls and electrical interfaces to note during mechanical inspections

Although this extract focuses on mechanical inspection, BS EN 17206 links mechanical safety to control-system safety. Mechanical inspection reports should therefore record where safety functions are present, missing, disabled, untestable or not evidenced.

Function / device	Inspection relevance
Control devices	Motion should be initiated and ended by a control device, direction clearly indicated, and movement generally maintained only while the operator actively enables the control device, e.g. deadman operation.
Emergency stop	Machinery installations should have emergency stop functions. Each operator control station needs an E-stop unless risk assessment justifies another arrangement. E-stop category is risk-assessment based.
Limit switches / overtravel	Where overtravel creates a hazard, initial and ultimate limits are required, or equivalent safety-related position sensing. Limits should remain active in normal operating modes, including maintenance.
Overload protection	Machinery movement should stop when load exceeds 1.2 times ELL. Elevators and accessible lighting bridges should not move in an overloaded condition.
Underload / slack protection	Where slack or underload can create a hazard, movement should stop when detected. This is especially important for suspended load bearing systems.
Improper winding	Where crossed grooves or rope doubling can create a hazard, movement should stop.
Overspeed	The control system should initiate a category 0 stop when operating speed exceeds 1.25 times rated speed.
Brake failure	Where brakes are safety devices, suitable measures should prevent damage or detect degradation. Regular inspection may be acceptable only where it demonstrably meets the required safety level.
Power source failure	If loss/reduction of supply or pressure can create a hazard, low-voltage or low-pressure protection should be provided.
Collision prevention	Where machines share a motion path, controls should prevent hazardous collision or other protective measures should be applied.
Group synchronisation	Where multiple machines travel interdependently and synchronisation loss increases risk, the group should stop when tolerances are exceeded. Tolerances should be documented.
Muting / override	Temporary overriding of safety functions is only for testing or special scenic situations, must not expose people to hazard, must be indicated, and should be removed after a set period/restart/user change.

7. Marking and documentation to check

7.1 Marking

Marking requirement	Inspection implication
ELL and ELL/R	Lifting machines should be marked with ELL and, where applicable, ELL/R. Where ELL varies by use case, the marking should make this clear.
Loading notices	Where load is not applied as a single vertical point load, notices should identify permitted load distributions. Notices should be prominent, for example on

	stage/studio floors, galleries, gantries, grid, anchorages and control points.
Supplementary loading information	Notices should explain counterweight storage loading, hoist/counterweight-set capacities, point vs distributed loading, where loads may be applied and any restrictions on permanently installed loads.
Machine rating label	Each machine should show manufacturer/importer, CE mark where relevant, type/series, serial number, year, ELL/ELL/R, intended use/use case, BS EN 17206 reference, power supplies and relevant speed/travel/control details.
Remote operation warning	Areas where remotely operated machines could start without warning should have clear warning signs.

7.2 Documents and records

Document / record	Why it matters
Technical data	Should include intended use, operating conditions, duty type, ELL/ELL/R, maximum point and area loads, travel speeds, acceleration/deceleration, travel path, synchronisation tolerances and limits.
Operating manual	Should explain machinery, intended use, forbidden use, operator competence, controls, monitoring, safeguards, maintenance, emergencies and replacement parts.
Installation instructions	Should cover handling, mounting, installation, commissioning and installation load testing, including load imposed at mounting points.
Repair and maintenance instructions	Maintenance should be by trained and competent persons. Routine maintenance requirements, frequency, lubrication, wear limits and spare parts should be specified.
Maintenance log	The manufacturer should provide a maintenance log. Inspectors should look for evidence of maintenance, faults and corrective action.
Inspection and examination guidance	Documentation should cover thorough examination after installation, including installation at a new site or location, before first service, and guidance for further inspection.
Test certificates	Copies of all relevant test certificates should be provided.
Technical file	The manufacturer should maintain a technical file for ten years from commissioning.

8. Testing prior to first use and after changes

8.1 Test sequence before first use

- Before first use, or after substantial changes, machinery should be tested in accordance with applicable national safety legislation and Clause 9 of the standard.
- If any test fails, corrective measures should be taken and the inspection/testing process repeated. The process should be fully documented and copies provided to all concerned parties.
- A test log should be compiled, consisting of manufacturer documentation and test reports.
- The test prior to first use comprises: preliminary test, structural/construction test, acceptance test and re-verification test if necessary.

Test stage	What should be checked
Preliminary test	Examination of technical documentation, including suspension mechanisms, load bearing equipment, hoisting accessories, drives, counterweights, travel ranges, safety equipment, structural calculations and diagrams.
Structural / construction test	Conformity with design/manufacturing documents, material requirements, crucial measurements, load bearing equipment and safety equipment arrangement, material certificates, rope/chain/pressure hose/gear certificates and welding qualification documents.
Acceptance test	Performed on equipment ready to operate, with operating instructions available. Assesses assembly, functional sequences, safety equipment, test loads and completeness of safety information/markings.
Re-verification	If defects are found, verification after rectification is limited to checking correction of those defects.

8.2 Safety equipment to test during acceptance

- Limit switches and travel limitation.
- Locking of stage lifts.

Stage Electrics - BS EN 17206:2020 Mechanical Inspection Extract

- Safety catches, pipe fracture safety devices and back-up safety nuts for spindle drives.
- Speed limiters.
- Load-limiting devices and pressure-limiting valves.
- Equipment preventing slackening of load bearing equipment.
- Synchronisation monitoring equipment.
- Emergency control elements and emergency stop functions.
- Safety circuits and continuous connection of the protective bonding conductor.
- Interlocking of command elements where there are multiple control points.
- Control-release equipment and signalling equipment.

8.3 Load testing requirements to flag

Application	Load test requirement / note
Loads at rest	Load tests should use at least 1.25 ELL/R.
Loads in motion	Load tests should use at least 1.10 ELL on downward travel, confirming lifting capacity and operation of brakes, clutches and hydraulic/pneumatic/electrical components.
Stage elevators with friction-locked load securing devices	Load tests should use 1.00 ELL/R.
Emergency stop function	Should be tested with ELL and at rated speed.
Friction-based holding devices without self-sustaining mechanism	Static load tests should be performed with ELL/R or by an equivalent method such as torque testing.
Positive holding devices / hydraulic cylinders on seat valves / locking mechanisms	Static tests with highest admissible standstill load are not necessary where the stated conditions apply.
Drive sheave powered lifting equipment	Test during travel with at least 1.3 times rated load and operational counterweight.
Jaw-operated winch drives	Test with at least 1.3 times rated load during travel and twice rated load during standstill.
Steel structures	Load tests of hoisting equipment test hoisting, coupling, braking and holding equipment. The steel structure itself does not require load testing where calculated proof is sufficient.
Travelling / turning equipment	For stage wagons and revolving stages, load tests are used to test drives and brakes. Structures may be justified by calculation. Direct manual auxiliary trucks do not require this load test.

8.4 Testing after changes and modifications

Change type	Inspection / test implication
Substantial changes	Further tests are required before operation resumes. Examples include changes to travel ranges/access points, control systems or SRP/CS, increased lifting capacity, drive/brake modifications, or design changes to supporting parts and load bearing equipment.
Large-scale repairs	Treated as substantial where they change machine design or replace parts with a different specification.
Like-for-like replacement	Replacing ropes, hydraulic cylinders, gears, motors or brakes with parts of the same specification is not classed as a substantial change.
Other changes	Tests should be performed before operation resumes, generally using acceptance-test principles. The examiner determines type and extent and may test individual components only.

9. Practical Stage Electrics inspection checklist

Inspection area	Questions to answer / evidence to record
Identification	What machine/system is being inspected? What is its use case, serial number, location, ELL/ELL/R and control arrangement?
Documentation	Are manuals, drawings, diagrams, certificates, maintenance logs, test logs and previous inspection/thorough examination records available?
Load path	Can the full load path be identified from load carrying device to anchor point? Are load bearing elements suitable, identifiable and secured against loosening?

Stage Electrics - BS EN 17206:2020 Mechanical Inspection Extract

Lifting accessories	Are hooks, shackles, slings, chains, ropes, beams or spreaders compatible, marked, undamaged, documented and appropriate for the load and use case?
Ropes/chains	Are ropes/chains correct type and condition? Are terminations compliant, inspectable, secure and free from unacceptable defects?
Drive and holding	Are brakes/load securing devices present, independent where required, testable individually and evidenced as effective?
Safety devices	Are limits, overload, underload/slack, overspeed, emergency stops, interlocks, anti-collision and synchronisation functions present and tested where required?
Safeguarding	Are crushing, shearing, trapping, fall, shaft, counterweight and moving-machine hazards controlled?
Access for inspection/maintenance	Are protective spaces, blocking devices, lighting, walkways, floor flaps and access controls suitable?
Markings and notices	Are ELL/ELL/R, loading notices, rating labels, warning signs and remote-operation signs present and legible?
Testing status	Has the installation been tested before first use or after changes? Is the acceptance report/test log complete?
Defects and actions	Are defects categorised, reported, rectified, retested and evidenced before return to service?

10. Important caveats

- This extract is a working summary for inspection planning and report drafting; it does not reproduce the full requirements of BS EN 17206.
- Always apply LOLER, PUWER, the manufacturer's instructions, venue-specific risk assessments, and any client or insurance requirements alongside this standard.
- For legal compliance, engineering judgement, unusual installations, performer flying or complex integrated systems, refer back to the complete standard and use a suitably competent person.

Source clauses reviewed

Primary source: BS EN 17206:2020+AC:2021, Entertainment technology - Machinery for stages and other production areas - Safety requirements and inspections. Key areas reviewed: Introduction; Clause 1 Scope; Clause 3 Terms and definitions; Clause 4 Hazards; Clause 5 Design requirements; Clause 6 Safeguarding hazardous areas; Clause 7 Electrical equipment and control systems; Clause 8 Documentation; Clause 9 Testing prior to first use and after substantial changes; Annexes A to G for hazard and application context.